

Presumptive Drug Identification Using Chemical Reagents (Field Procedure)



1. Purpose

This Standard Operating Procedure (SOP) defines the standardized method for conducting presumptive drug identification using chemical colorimetric reagents under field conditions without laboratory equipment.

2. Scope

This procedure applies to:

- Border Guard forensic technicians in training,
- roadside inspections,
- preliminary field investigations,
- controlled training environments.

NOTE: Results obtained are presumptive only and require confirmatory laboratory analysis.

3. Definitions

- **Presumptive Test** – A rapid chemical test indicating the possible presence of a controlled substance.
- **Reagent** – A chemical solution that produces a characteristic color change when in contact with specific drug classes.
- **Sample** – Unknown material collected for testing (powder, tablet, plant material).

4. Responsibilities

- The operator is responsible for performing the test safely, documenting results, and ensuring proper disposal.
- Supervisors must ensure technicians are trained and reagents are within expiration dates.
- All personnel must comply with safety and evidence-handling protocols.

5. Safety Requirements

WARNING: Chemical reagents may cause burns, irritation, or toxic fumes. Handle with extreme caution.

Personal protective equipment (PPE):

- Nitrile gloves
- Safety goggles
- Protective apron

Environmental conditions:

- Perform tests outdoors or in a well-ventilated area.
- Keep reagents away from heat and direct sunlight.

Handling precautions:

- Avoid skin and eye contact.
- Do not inhale fumes.
- Do not mix reagents unless specified.

Waste disposal:

- Follow hazardous waste protocols.
- Neutralize reagents before disposal.

6. Required Materials

- Transparent glass vials or test tubes with stoppers
- Glass droppers or pipettes
- White ceramic or glass plate (color comparison)
- Field reagent kit (Marquis, Mandelin, Mecke, Scott, Duquenois–Levine, Simon's)
- Waste container for used reagents and samples

7. Procedure

7.1 Sample Preparation

1. Place a small amount of the unknown substance (match $\square$ head size) into a clean, dry vial.
2. If the sample is a tablet or plant material, gently crush it using a clean glass rod.
3. Do not add water or neutral liquids. Reagents act as the reaction medium.

7.2 Reagent Selection

Select the reagent based on the suspected drug class:

Reagent	Target Substances	Expected Color Change
Marquis	Opiates, amphetamines, MDMA	Purple (opiates), orange-brown (amphetamines)
Mandelin	Methamphetamine, ketamine, MDMA	Green (meth), black (MDMA)
Mecke	Heroin, morphine	Blue-green $\rightarrow$ black
Scott (3-step)	Cocaine	Blue $\rightarrow$ Pink $\rightarrow$ Blue
Duquenois–Levine	Cannabis	Purple in chloroform layer
Simon's	Secondary amines (MDMA)	Blue

7.3 Testing Procedure

WARNING: Do not lean directly over the vial. Some reactions release irritant vapors.

1. Add **1–2 drops** of the selected reagent onto the sample.
2. Observe the reaction for **up to 60 seconds**.
3. Compare the color change to the reference chart.

8. Interpretation and Documentation

Record the following information:

- Substance tested
- Reagent used
- Time to color change
- Final color observed
- Reagent batch/lot number
- Operator initials
- Location and date

NOTE: All results must be marked as *presumptive*.

9. Waste Handling and Decontamination

- Neutralize used reagents with a baking soda solution.
- Dispose of neutralized waste in designated hazardous waste containers.
- Rinse all glassware with distilled water and allow to air dry.
- Store reagents in sealed, labeled containers.

10. Training Notes

- Include simulated samples and known false positives.
- Reinforce limitations of field testing.
- Emphasize consistent documentation and safety compliance.
- Encourage critical thinking and avoidance of confirmation bias.